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Architecture and data flow of the circle-packing classroom

One loopback Python server serves every teaching view and owns a single active packing run. The problem is AlphaEvolve's unit-square sum-of-radii task for N=26 or N=32. The model is TAMU Luna. Our local controller and initial numerical methods are classroom implementations, not the unpublished AlphaEvolve engine.

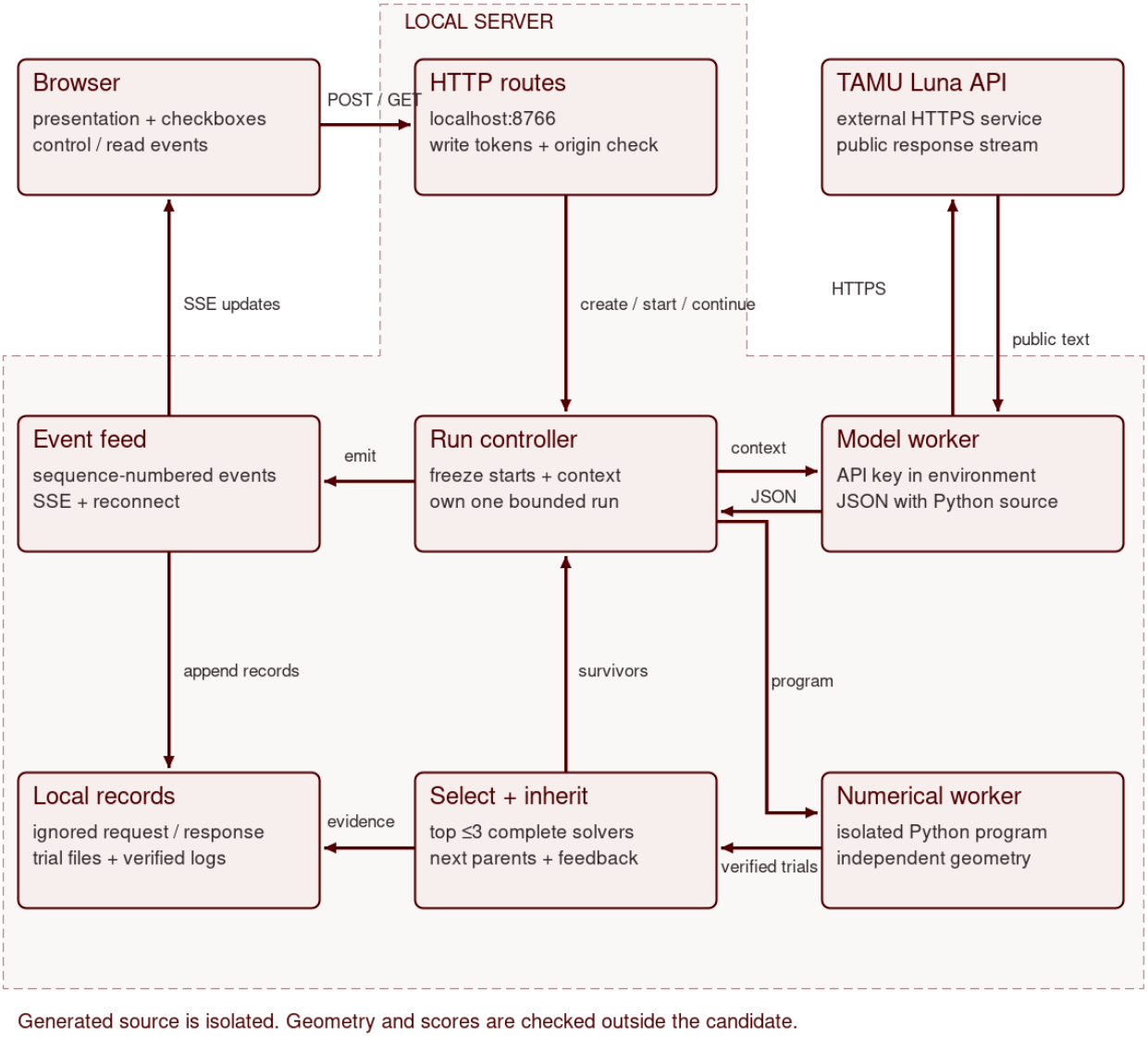


Figure 1: Server, model worker, isolated numerical execution and checked feedback.

Search step, trial and evolutionary round

A search step proposes a center change, reallocates valid radii, and applies a move acceptance rule. In Greedy Local Center Search, a move survives only when the sum of radii increases. The move scale controls the typical random coordinate change. It is not a fixed or maximum distance.

A trial executes one unchanged program once: solve(initial, seed, steps). It receives one starting-center array, one numerical random seed, and an 80-step work request, with an eight-second hard wall-time cap. It can produce many intermediate packings before returning one final packing. With default seeds, trials 1, 2 and 3 use 3, 4 and 5. Every competing program receives copies of the same three starting packings, called the packing bank, with corresponding seeds and limits. The requested step count is not a guarantee of actual work.

An evolutionary round compares parent programs, requests one new source program from Luna, evaluates it and selects survivors. A trial is numerical execution, not a model request. A full run includes initial evaluation and up to three such rounds. For three complete, valid trial scores S1, S2 and S3, the displayed

primary fitness is F=median(S1,S2,S3), maximized. Chapter 10's minimized convention uses f=-F. Any invalid or incomplete trial makes the program ineligible.

Ownership and modules

Module	Responsibility
server.py	Loopback HTTP, tokens/origin checks, routes, SSE and presentation preview
packing_controller.py	Run state, programs, banks, trials, requests, selection, Stop and replay
packing.py	Mathematical contract, seeded starts, source identity, independent geometry/score
packing_profile.json	Circle counts, trial/call/run limits and numerical-mode defaults
packing_programs/	Three disclosed initial source programs
packing_sandbox.py	Mandatory Bubblewrap runtime isolation and availability check
packing_worker.py	Execute solve; provide fixed-center radius LP and progress reporter
model_worker.py	One bounded tool-free provider request; project public response stream
settings.py	Read ignored local configuration; bind credentials to the correct provider
static/packing.*	Controls, source/diff, trial table, packing view, exchange and evidence
build_presentation.py / preview.py	Shared prose to frames/notes; saved-source preview

Request and return

The browser creates a run, then starts numerical evaluation. After parents have been measured, each model round waits at an explicit gate. The request contains task/interface/limits, parent source, current-bank scores, failed and successful history, tried IDs, numerical seeds and remaining time. Parent source is sorted by stable ID so structure-only comparisons do not leak fitness through ordering.

The model worker sends HTTPS to the configured TAMU endpoint with the private Authorization header, system instructions/schema and serialized context. It has no execution tools. Public content streams to the browser; hidden reasoning and the header are not stored as teaching evidence. No API key enters the prompt.

The JSON response contains round_id, parent_ids, hypothesis and source. Source must define solve(initial,seed,steps). The controller validates fields, ancestry, syntax, signature and normalized duplicate identity before admission. The source string implements the proposal; the hypothesis does not tell Python what to do.

Execution and the trust boundary

A separate owned process invokes Bubblewrap, unsharing network, process and mount namespaces. It exposes read-only Python/runtime libraries and the trusted worker, not the repository, private home, .env, run records or host network. It receives only source, starts, seed, circle count and work request through stdin. Environment credentials are absent. Missing isolation blocks execution; there is no fallback that executes generated source inside the server.

The worker supplies NumPy, math, radii and report. For fixed centers, the radius subproblem is linear: maximize summed radii under pairwise distance constraints and wall upper bounds. SciPy HiGHS solves it with a common construction margin. The candidate can implement center search in actual Python. CPU/memory/file limits, an owned wall deadline and bounded output capture limit execution. Syntax checks are contract checks, not the security boundary.

Every displayed progress witness and final result is checked again in the server: exactly N finite triples, nonnegative radii, all walls and all pairs. Recompute summed radii independently. Do not trust a candidate's own objective or success flag. Tolerance is 1e-9; the allocator's common construction margin is 1e-8. Floating checks are not formal certificates. A progress witness can be valid while its program later fails; it supplies no completed-trial fitness by itself.

Packing banks and selection

Initialize from common seeded perturbed grids. Default iterative refinement keeps the best valid packing per trial seed, freezes those centers for the next round, then re-evaluates parents and child on copies of that same bank. Every evaluation records the bank identity. Only comparable complete programs enter selection. The optional fixed protocol keeps the original bank for all rounds.

All three trials must complete and validate. Rank by median sum, minimum sum and stable source identity; retain the top three. This implements elitist pooling, not islands or tournament sampling. Syntax-distinct programs need not be diverse. Keep the cumulative best verified construction separately from program fitness; old-bank medians must not be treated as a stable program-quality time series.

Routes, lifecycle and evidence

Route	Purpose
/	Presentation hub, demo transition and return
/packing or /demo	Active circle-packing experiment
/algorithms or /packing-guide	Mechanisms and worked examples
/exchange	Demo with exact exchange panels
/notes, /handout.pdf, /document.pdf	Shared teaching content and exports
/architecture, /architecture.pdf	This document
/api/packing/status	Safe configuration, programs, genuine recordings and write token
POST /api/packing/runs	Create; no numerical/model execution yet
POST /api/packing/runs/ID/start	Begin bounded run and initial numerical evaluations
POST …/continue	Release one pending model gate; duplicate action is idempotent
POST …/stop	Stop owned work
POST /api/packing/reset	Stop existing work and create a fresh run

Route	Purpose
GET …/events	SSE with sequence cursor; reconnect never repeats a call
GET …/exchange/ROUND	Actual request body and public response; no Authorization

Per trial: 80 requested search steps and an eight-second hard wall cap, including startup. Per model call: 35 seconds. Per run: at most three calls and 240 seconds, with bounded owned cleanup. Faster runs leave teaching time for inspection; no artificial wait is inserted. A cap is not a promise of every algorithm completing.

Each run has an ignored folder with versioned task, source, events and actual request/public return. LIVE LLM uses a real provider; NUMERICAL ONLY never calls one; RECORDED REPLAY requires real source provenance and checked geometry. It reveals public responses only on the replay timeline. A failure remains visible and never causes an unannounced retry or fabricated substitute.

The browser holds write tokens in its same-origin session. Host/origin validation and tokens protect mutations. Static routes are explicitly allowlisted. No route serves .env, arbitrary repository paths or raw private run directories. Another account must supply its own ignored credentials after cloning.

Post-search original-bank comparison

After a live or numerical run finishes, the explicit comparison action freezes the initial programs and final survivors. PackingManager.validate_original creates a separate NUMERICAL ONLY task with original centers, the same three seeds and numerical limits, and a fresh 240-second task deadline. Identical source is evaluated once and displayed in each applicable table: initial programs and final survivors. The parent run exposes a separate comparison snapshot; its evolution population, fitness history and construction curve remain unchanged. Stop/Reset cancels owned comparison workers. Repeated comparison requests return the same task rather than launching another evaluation. No provider request or evolutionary feedback occurs. Original starts are familiar conditions, not a held-out test.